Lab Assignment 7: MOSFET Circuits
Revision: February 27, 2023

Summary

Metal oxide semiconductor field effect transistors (MOSFETs) are the driving force behind advancement in technology for the past 40 years. In this lab, you will experimentally obtain I-V characteristic curves for two types of the MOSFET transistors: NMOS and PMOS. You will use the TEK571 Curve Tracer to obtain the I-V curves for the NMOS ZVN2110A and the PMOS ZVP2110A. Once the curves are obtained, the you will be able to estimate essential parameters (k_n, k_p, V_t, g_m and λ) from the curves. Then, you will use the ZVN2110A NMOS as a voltage controlled variable resistance in the deep triode region. Finally, you will be able to use the IRF510 NMOS power transistor as a power amplifier. The power amplifier receives a weak power signal and then amplifies its power at the output.

Learning Outcomes: After completing this lab, you should be able to:

- Understand the current-voltage equations for the NMOS and PMOS transistors.
- Experimentally obtain the I-V curves for the PMOS and NMOS transistors.
- Experimentally estimate essential parameters k_n, k_p, V_t, g_m and λ from the I-V curve.
- Design, build and test a MOSFET as a voltage controlled variable resistance in the deep triode region.
- Design, build and test a MOSFET transistor as power amplifier.

Required Equipment

- EE352 Analog parts kit
- Breadboard
- Function Generator
- Oscilloscope
- DMM (2)
- DC Power Supply
- TEK 571 Curve tracer or Discovery III

I. MOSFET I-V Characteristics

In this part, you will use the TEK571 curve tracer shown in Figure B1 in Appendix B of this lab assignment to obtain I-V curves for the NMOS ZVN2110A and the PMOS ZVP2110A transistors. You may use Digilent Discovery III board if a curve tracer is not available. You will obtain I-V graphs for each transistor, including an I_D vs V_{GS} curve, and a family of I_D vs V_{DS} curves at multiple steps of V_{GS}. Then you will use the curves to experimentally calculate the parameters (k_n, k_p, V_t, g_m and λ).

Pre-lab:

(a) Read Sedra and Smith (7th edition) section 5.2 and refer to Figures 5.14 on pp. 268 and Figure 5.17 on pp. 272.

(b) Write down the I-V equations for NMOS and PMOS transistors in cutoff, triode and saturation region and state the conditions for each region.

(c) The transconductance g_m in the saturation region is defined as

$$g_m = \left.\frac{di_D}{dv_{GS}}\right|_{I_D}$$

Write down another two equivalent expressions for the transconductance (g_m).

(d) From the datasheet of the ZVN2110A and ZVP2110A find the range of the threshold voltage for both transistors.

Lab Procedures:

A. *Using the TEK571, obtain the I_D vs V_{GS} curve for the ZVN2110A NMOS transistor.*

1. You need to setup the TEK571 curve tracer to obtain the I_D vs V_{GS} curve. From it, you will be able to experimentally estimate the threshold voltage V_t and the transconductance (g_m). You may use Digilent Discovery III board if a curve tracer is not available.

2. Connect the transistor in a diode connected configuration as shown in Figure B2, and its equivalent circuit shown in Figure B3 in the appendix of this handout.

3. Click on the Menu button and then navigate through the setup and select the following: Function=Acquisition, Type=NFET, Vds max = 5 V, Is MAX = 10 mA, Vg/step = 0.2, Offset = 0, Steps = 1, Rload=0.25 Ohms, p max = 0.5 Watt. Then Click on Start. Use similar setting if you are using Digilent Discovery III board.

4. You should get a diode curve that plots I_D vs. V_{GS}. From the curve obtain the threshold voltage, V_t, which is the minimum voltage to that is required to create the channel so that the current starts conducting. You need the value of V_t in next step. Take a clean and straight picture of the curve using your smartphone camera. Make sure that your hand is still and the picture is focused.

B. *Using the TEK 571, obtain I_D vs V_{DS} curves at multiple steps of V_{GS} for the ZVN2110A NMOS transistor*
   1. Connect the transistor in its normal configuration by returning the gate to its corresponding pin of the same socket as in Step A.2.
   2. You need to setup the curve tracer to plot the I-V curves for the transistor using multiple steps of $V_{GS.}$ From the I-V curves, you will be able to obtain k_n and λ for the transistor from those curves.
   3. Click on the Menu button and then navigate through the setup and select the following: Function=Acquisition, Type=NFET, Vds max = 10 V, Is MAX = 50 mA, Vg/step = 0.1, Offset = V_t+0.1, Steps = 5, Rload=0.25 Ohms, p max = 0.5 Watt. Then Click on Start.
   4. You will have six curves. The first curve is when V_{GS}=OFFSET. The second curve is when V_{GS}=OFFSET +0.1, the third curve is when V_{GS}=OFFSET+2*(0.1), and so on. Hence, you have a family of I_D vs V_{DS} for different values of $V_{GS.}$
   5. Take a clean and straight picture of the curve using your smartphone camera. Make sure that your hand is still and the picture is focused.

C. *Using the TEK571, obtain the I_D vs V_{SG} curve for the ZVP2110A PMOS transistor.*
   1. Basically, you will have to repeat what you have done for the NMOS transistor. Connect the transistor in a diode connected configuration as shown in Figure B2.
   2. Click on the Menu button and then navigate through the setup and select the following: Function = Acquisition, Type=PFET, Vds max = 5 V, Is MAX = 10 mA, Vg/step = 0.2, Offset = 0, Steps = 1, Rload=0.25 Ohms, p max = 0.5 Watt. Then Click on Start.
   3. You should get a diode curve that plots I_D vs. V_{SG}. From the curve obtain the negative value of threshold voltage,-V_t. Take a clean and straight picture of the curve using your smartphone camera. Make sure that your hand is still and the picture is focused.

D. *Using the TEK 571, obtain I_D vs V_{DS} curves at multiple steps of V_{SG} for theZVP2110A PMOS transistor*
   1. Connect the transistor in its normal configuration by returning the gate to its corresponding pin of the same socket.
   2. You need to setup the curve tracer to plot the I-V curves for the transistor using multiple steps of $V_{SG.}$ From the I-V curves, you will be able to obtain k_n and λ for the transistor.
   3. Click on the Menu button and then navigate through the setup and select the following: Function=Acquisition, Type=PFET, Vds max = 10 V, Is MAX = 50 mA, Vg/step = 0.2, Offset = -V_t-0.1, Steps = 5, Rload=0.25 Ohms, p max = 0.5 Watt. Note that the offset is a negative value because the PMOS requires a negative V_{GS} to turn it on. Then Click on Start.

4. You will have six curves. The first curve is when V_{GS}=OFFSET. The second curve is when V_{GS}=OFFSET - step, the third curve is when V_{GS}=OFFSET - 2* step, and so on. Hence, you have a family of I_D vs V_{DS} for different values of V_{GS}.

5. Take a clean and straight picture of the curve using your smartphone camera. Make sure that your hand is still and the picture is focused.

E. Estimating the parameters from the obtained graphs.

1. Port your curve tracer pictures into your PC and print them or place them within a MS Word document. They can be printed and attached to your report later.

2. For the NMOS and PMOS transistors and from the I_D-V_{GS} curves, estimate the value of the threshold voltages which are the minimum voltage required to start conducting current. Note that V_{tn} >0 but V_{tp} <0. Record those values in your laboratory notebook.

3. The transconductance g_m in the saturation region is defined as

$$g_m = \left.\frac{di_D}{dv_{GS}}\right|_{I_D} \qquad (1)$$

   which is the slope of tangent to the curve at a specific current (I_D). Note that the slope of the tangent varies by varying I_D. Pick a point from the I_D-V_{GS} curve for the NMOS and estimate its transconductance by drawing that tangent at that point and then estimating the slope. Using your measured g_m from the slope, estimate the k_n from the two other equations for g_m. Record all values on your lab notebook.

4. Repeat step 3 for the PMOS transistor.

5. Estimate the channel modulation index λ for both transistors and record the values for each transistor.

6. For the NMOS I-V_{DS} family of curves, pick 6 different points on the curve (three points in the saturation region and three in the triode region) and record them in a table like the one shown. Then estimate their process transconductance (k_n). Then calculate the average value of k_n. Comment on your results.

I_D	V_{GS}	V_{DS}	Triode/Saturated	Calculated k_n

7. Repeat step 6 for the PMOS transistor.

F. *Show the TA your results for kn and kp and your graphs and discuss your results with him/her. Then have him/her initial your checklist.*

Post Lab Assignment

Using the estimated parameters from Step 5, use PSPICE (LTSPICE or Orcad) to simulate i_D vs. v_{DS} characteristics for both the n- and p-channel MOSFETs. Compare your simulated and measured characteristic curves.

II. FET as Voltage Controlled Resistor

From part I, you observed that the I_D-V_{DS} relationship is linear in the deep triode region. Hence, the drain-source channel of the transistor acts as voltage controlled resistor that depends on V_{GS}. The slope of the I_D-V_{DS} relationship varies by changing V_{GS}. You can easily derive the resistance value from the I-V equation in the deep triode region, which is

$$I_D = k_n\left[(V_{GS} - V_t)V_{DS} - \frac{1}{2}V_{DS}^2\right]. \qquad (2)$$

Since $V_{DS} << V_{GS}$ in the deep triode region then

$$I_D \approx k_n(V_{GS} - V_t)V_{DS}. \qquad (3)$$

Now, the drain to source resistance (R_{DS}) can be expressed as

$$R_{DS} \approx \frac{V_{DS}}{I_D} = \frac{1}{k_n(V_{GS} - V_t)}. \qquad (4)$$

Hence, the transistor in the deep triode region acts as a voltage controlled resistance that is controlled by varying V_{GS}, and the transistor can be replaced with a variable resistance as shown in Figure 1.

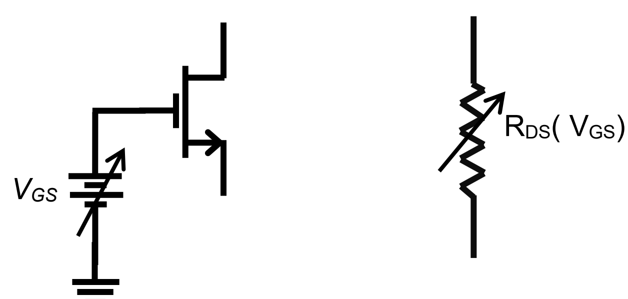

Figure 1. NMOS as a variable resistance in the deep triode region.

Pre-lab:

(a) For the voltage divider circuit shown in Figure 2, the transistor is acting as a variable load resistance in the deep triode region. Based on the parameter values that you have obtained for the ZVN2110A NMOS transistor in part I, design the circuit by choosing R such that V_o will change from 0.1 V to 0.02 V such that $V_{GS} \leq 5V$. Then calculate V_{GS} and r_{DS} that would lead to V_o = 0.1V and 0.02 V.

Lab Procedures:

1. Construct the circuit of Figure 2 using the same ZVN2110A NMOS transistor that was characterized in Part I and using your designed value for R.
2. Using DC power supply, apply the calculated V_{GS} values in the pre-lab to obtain V_o = 0.1 V and V_o = 0.02 V. Use DMM to measure V_{GS} values and their corresponding V_o values. Then estimate the error in V_o at both ends. Record your values in your lab notebook. A large error would be caused by inaccurate estimated values for k_n and V_t from part I.
3. Vary V_{GS} until V_o = 0.1 V and record its value, then vary V_{GS} until V_o = 0.02 V. Estimate the error between expected V_{GS} and measured V_{GS}. Record your measured values in your lab notebook.
4. *Show your circuit and your measured values to your TA and then have him or her initial your checklist.*

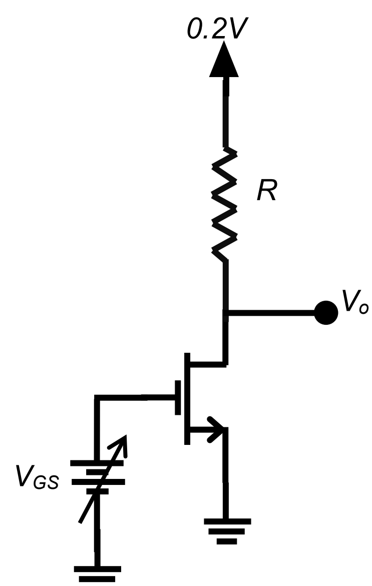

Figure 2. NMOS as a variable resistance in a voltage divider circuit.

III. **MOSFET as Buffer Amplfier**

A buffer amplifier is used when the input signal source is not capable of supplying the load with large current, therefore power, while maintaining the same voltage level as the input. This arises in many practical situations in electronics such as in sending am audio signal to a speaker, where you want the output voltage to be equal to the input voltage but you want the current, or power, to be much larger than what the input signal source can handle. Since the current greatly increases and voltage remains the same, you would expect the output power to also greatly increase. The circuit of Figure 3 is called a Buffer Amplifier (or Common Drain amplifier), which is a power amplifier. In this part, you will investigate the large signal transfer characteristics from the input to the output of the amplifier by varying the input voltage from 0 to V_{DD}.

Pre-lab:

(a) For the circuit in Fig. 3, assume V_i will be varied from 0 to V_{DD} (15 V), and calculate the range of the power which is supplied by the input source $V_i = V_G$.

(b) For the circuit in Fig. 3, assume that V_o will range from 0 to V_{DD} (15 V), and calculate the range of power which can be delivered to the load.

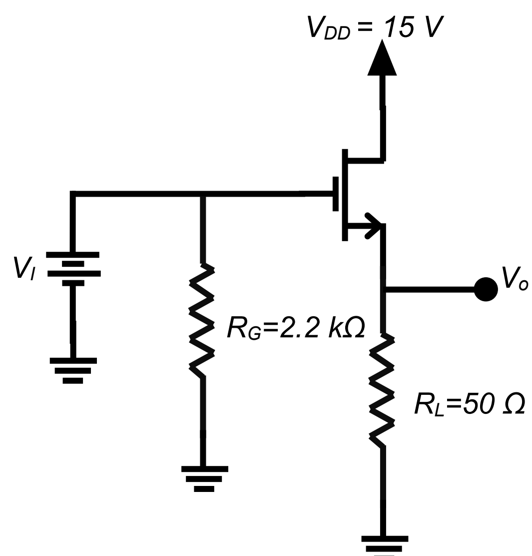

Figure 3. NMOS buffer amplifier.

Lab Procedures:

1. Using IRF510 NMOS power transistor, connect the circuit of Figure 3. Figure A2 in Appendix A shows the pin assignments for IRF510 transistor. Use one of the decade box power resistors that we have in the lab for R_L, as the resistors that you have in the parts kit can only handle up to 0.25W. Measure and record the resistor values in your lab notebook. Use one DC power supply to supply V_{DD}=15 V, and the other DC power supply to be applied at the gate of the amplifier.

2. Use DMM1 & DMM2 to measure V_I and V_o simultaneously. Determine the threshold voltage of the transistor by starting with V_I from zero and then slowly increasing V_I until V_o departs from zero. Record the threshold voltage of the transistor in your lab notebook.

3. Starting from zero vary V_I in increments of 1 volt all the way 15 volts and measure V_o accordingly. Record your data in a Table, and then calculate the power supplied by the input P_i, the power delivered to the load P_L and the region of operation of the transistor (triode, saturation or cutoff).

4. Enter the data into a MS excel or Matlab and plot V_o vs V_i and P_o vs P_i.

5. *Show your circuit and your measured values to your TA and then have him/her initial your checklist.*

Lab Report:

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet with Laboratory Instructor's initials indicating completed lab demos to your report.

Appendix A: Components Packages and Pins Assignments

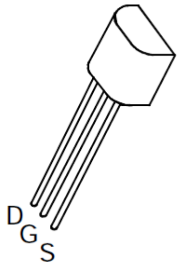

Figure A1. TO-92 Package for ZVN2210A and ZVP2110A and their pin assignments.

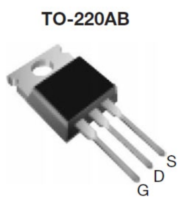

Figure A2. TO-220AB Package for IRF510 power NMOS transistor.

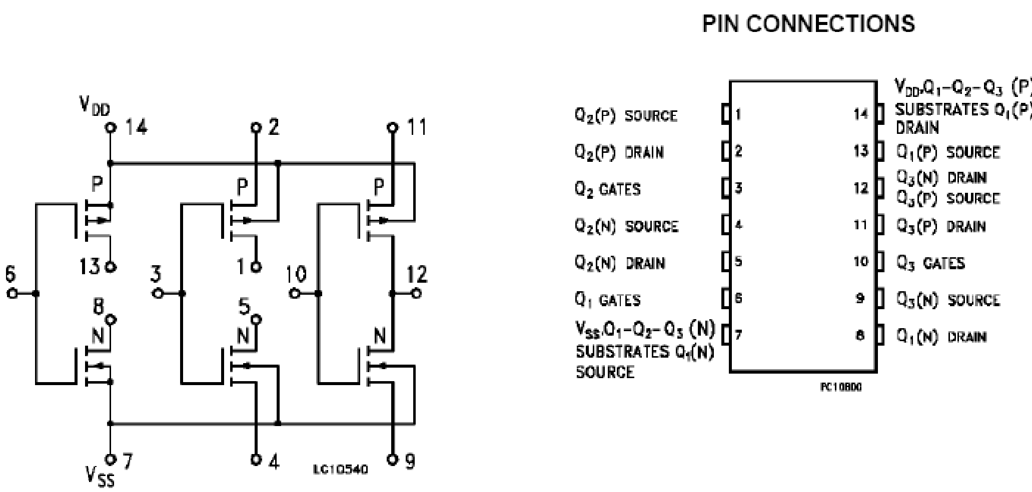

Figure A3. Package and pin assignment for CD4007 complementary dual CMOS transistors.

Appendix B TEK 571 Curve Tracer

Figure B1. TEK 571 Curve Tracer.

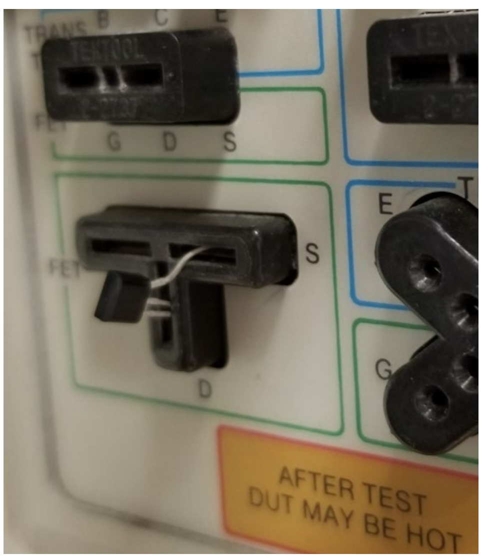

Figure B2. Diode connected FET.
Note: Use FET socket.
Shorted drain/gate does in D socket, source goes in S socket, for both NFET and PFET.

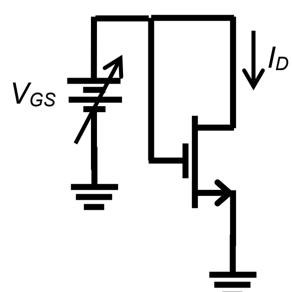

Figure B3. Diode connected circuit of figure B2.

Lab 7 Checklist

Name:__________________

Lab title and introduction (5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (two or three sentences) explaining the purpose of this lab.

I. MOSFET Characteristics (45 pts total)

1. Printout of measured NMOS I-V curves (3 pts)
2. Printout/sketch of measured PMOS I-V curves (3 pts)
3. Calculated values of V_T, K_n, λ, and g_m for the NMOS (10 pts)
4. Calculated values of V_T, K_P, λ, and g_m for the PMOS (10 pts)
5. ________ **DEMO:** Have a TA initial this sheet, indicating that he/she has observed your calculations of the NMOS parameters. (5 pts)
6. PSPICE simulation to get the I-V curves for NMOS using values from 3. (5 pts)
7. PSPICE simulation to get the I-V curves for PMOS using values from 4. (5 pts)
8. Compare simulated vs. measured (I-V) curves for both NMOS and PMOS. Comment on differences. (4 pts)

II. Voltage Controlled Resistor (25 pts total)

1. Circuit diagram. (2 pts)
2. Explanation of how you computed R_{DS} for a given V_{GS}. (3 pts)
3. Designed and measured values of R. (5 pts)
4. Estimated and measured values of V_o at 10% and 50% of V_{DD}. (5 pts)
5. Estimated and measured values of V_{GS} at 10% and 50% of V_{DD}. (5 pts)
6. ________ **DEMO**: Have a TA initial this sheet, indicating that they he/she has observed your circuit's operation. (5 pts)

III. MOSFET Buffer Amplifier (25 pts total)

1. Circuit diagram used to measure voltage transfer characteristic with measured values for R_G and R_L. (3 pts)
2. Determine the threshold voltage of the IRF510 transistor. (3 pts)
3. Plot of voltage transfer characteristics, V_o vs. V_I. (6 pts)
4. Plot of output vs. input power transfer characteristics. (6 pts)
5. ________ **DEMO**: Have a TA initial this sheet, indicating that he/she has observed your circuit's operation. (7 pts)
   - Show TA your plot of the system power transfer characteristic.

